

GenAI Website Builder — Hackathon Brief

Individual · 6 Hours · Deliverable: One live, working public URL

The Challenge

Build a chat interface where a user describes a website and the system generates and renders it live in the browser — instantly.

The loop: Prompt in → LLM generates code → site renders → user refines via chat.

Output Format — Your Choice

HTML/CSS/JS rendered in an iframe, or **React components** rendered dynamically. Both are valid. Judges score output quality, not format. Pick what you ship fastest in.

Feature Tiers

Tier 1 — Must Ship (*by Hour 3*)

- Chat input → Groq LLM → live site preview rendering in the browser
- At least 3 prompt types work well (landing page, portfolio, form page)
- App deployed to a public URL

Tier 2 — Strong Submission (*by Hour 5*)

- Follow-up prompts refine the existing site, not regenerate from scratch
- Structured JSON output { `html`, `css`, `js`, `title` } validated with Pydantic
- Graceful error handling for malformed LLM output
- Langfuse tracing on every LLM call

Tier 3 — Outstanding (*Hours 6*)

- Style controls — user picks palette, font, or layout before generating
- Version history — revert to any previous generated version

- Multimodal — upload a sketch or screenshot, generate a site from it
 - RAG-powered design inspiration injected into the system prompt
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Judging Rubric

Criterion	Weight	What Judges Look For
UI/UX of Generated Sites	40%	Do outputs look good and work? Quality consistent across prompt types?
Prompt-to-Output Accuracy	35%	Does output match the ask? Does refinement work? Edge cases handled?
Code Quality & Architecture	25%	Clean code, Pydantic validation, error handling, Langfuse wired in

Judges will **test 5 live prompts** on your URL. A polished wrapper on a broken generator scores low. The generated site is the product.

Timeline

Hour	Focus	Milestone
1	Setup, first LLM call working	LLM responds to a prompt
2–3	Core loop + preview rendering	Site renders from a prompt
4	Deploy to live URL	 Public URL live
5–6	Tier 2 features + edge case testing	Iterative editing works
7–8	Tier 3 / polish / demo prep	Submission ready

Deploy at Hour 3, not Hour 6. Deployment issues need time to fix.

Submission Checklist

- [] **Live URL** — publicly accessible, tested before submitting
 - [] **GitHub repo** — README with local setup in under 5 steps
 - [] **5-min demo** — best prompt · architecture · what you'd build next
 - [] **One-paragraph reflection** — what broke, what surprised you
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Rules

- Individual submission — no teams
- You may use: Agno, Groq, any open-source model, any UI library
- You may not use: pre-built website builder APIs (Webflow, Wix, etc.) — the LLM must do the generating
- API keys go in environment variables — never in the repo